

Histopathology Classification and Explanation of Lung and Colon Cancer Using Graph Neural Networks

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Abstract

Histopathological image analysis plays a crucial role in the diagnosis and treatment planning of cancer. Traditional approaches for histopathology classification often rely on convolutional neural networks (CNNs) that may not fully capture the complex spatial relationships within tissue structures. In this paper, we propose a novel Graph Neural Network (GNN) approach for histopathology classification and explanation of lung and colon cancer using the LC25000 dataset. Our approach dynamically constructs graphs from image features and employs Graph Attention Networks (GATv2) for classification. The model achieves state-of-the-art performance with 99.8% accuracy and 0.99999 AUC across the five classes in the dataset. Furthermore, we introduce a comprehensive explanation framework that combines GNN-based explanations, feature importance visualization, and LIME-based superpixel explanations to provide interpretable results. Experimental results demonstrate that our approach not only outperforms existing methods in terms of accuracy but also offers superior interpretability, which is crucial for clinical applications. The proposed method represents a significant advancement in the application of graph-based deep learning for histopathology image analysis.

1 Introduction

Cancer remains one of the leading causes of mortality worldwide, with lung and colon cancers among the most prevalent and deadly forms. According to the World Health Organization, lung cancer accounts for approximately 2.2 million new cases and 1.8 million deaths annually, while colorectal cancer contributes about 1.9 million new cases and 935,000 deaths globally. Early and accurate diagnosis is crucial for improving patient outcomes, with histopathological examination of tissue samples serving as the gold standard for cancer diagnosis.

Histopathology involves the microscopic examination of tissue samples to study disease manifestations, particularly for cancer diagnosis and staging. Traditional histopathological analysis relies on pathologists manually examining tissue slides under a microscope, a process that is time-consuming, labor-intensive, and subject to inter-observer variability. The increasing incidence of cancer, coupled with a shortage of pathologists, especially in developing regions, has created an urgent need for automated and reliable diagnostic tools.

In recent years, deep learning approaches have shown remarkable success in medical image analysis, including histopathology image classification. Convolutional Neural Networks (CNNs) have been the predominant architecture used for this task, achieving high accuracy in classifying various types of cancer from histopathological images. However, CNNs primarily focus on local patterns and may not fully capture the complex spatial relationships and tissue architecture that are crucial for accurate cancer diagnosis.

Graph Neural Networks (GNNs) have emerged as a promising alternative for histopathology image analysis due to their ability to model relationships between different regions of an image. By representing an image as a graph, where nodes correspond to regions or cells and edges represent their spatial or feature relationships, GNNs can better capture the structural information that is essential for histopathological diagnosis. Despite their potential, the application of GNNs to histopathology image classification remains relatively unexplored compared to CNN-based approaches.

In this paper, we propose a novel GNN-based approach for histopathology classification and explanation of lung and colon cancer using the LC25000 dataset. Our approach makes several key contributions:

1. We introduce a dynamic graph construction method that learns to create edges between image regions based on feature similarity, allowing the model to adapt to the specific characteristics of each image
2. We employ Graph Attention Networks v2 (GATv2) for message passing between nodes, enabling the model to focus on the most relevant parts of the graph for classification
3. We develop a comprehensive explanation framework that combines GNN-based explanations, feature importance visualization, and LIME-based superpixel explanations to provide interpretable results
4. We demonstrate state-of-the-art performance on the LC25000 dataset, achieving 99.8% accuracy and 0.99999 AUC across the five classes (colon adenocarcinoma,

benign colonic tissue, lung adenocarcinoma, lung squamous cell carcinoma, and benign lung tissue)

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 describes the dataset, Section 4 presents our methodology, Section 5 details the explainability framework, Section 6 shows results, Section 7 discusses findings, and Section 8 concludes.

2 Related Work

2.1 Histopathology Image Classification

Histopathology image classification has seen significant advancements with the application of deep learning techniques. The LC25000 dataset (Borkowski et al. 2019), has become a benchmark for evaluating methods for lung and colon cancer classification. This dataset contains 25,000 histopathological images across five classes: colon adenocarcinoma, benign colonic tissue, lung adenocarcinoma, lung squamous cell carcinoma, and benign lung tissue.

Early approaches to histopathology image classification primarily relied on traditional machine learning methods with handcrafted features. These methods typically involved extracting texture, color, and morphological features followed by classification using algorithms such as Support Vector Machines (SVMs) or Random Forests (Sirinukunwattana et al. 2016). While these approaches provided interpretable results, they often failed to capture the complex patterns present in histopathological images.

With the rise of deep learning, Convolutional Neural Networks (CNNs) have become the dominant approach for histopathology image classification. Hadiyoso et al (Hadiyoso et al. 2021) proposed a deep learning technique for automatically classifying colon and lung cancer using the LC25000 dataset, achieving high accuracy but with limited interpretability. Transfer learning approaches using pre-trained models such as VGG-19, ResNet, and EfficientNet have been particularly successful. Sanidhya et al. (Sanidhya et al. 2022) developed a CNN pre-trained diagnostic network for lung and colon cancer, achieving 96% and 97% accuracy for colon and lung cancer classification, respectively. More recently, ensemble and fusion methods have shown promising results. Abd El-Aziz et al. (Abd El-Aziz et al. 2024) proposed an advanced deep learning fusion model that integrates features from ResNet-101V2, NASNet-Mobile, and EfficientNet-B0 for the multi-classification of lung and colon cancers. Their model achieved impressive results: 99.8% precision, 99.8% recall, 99.8% F1-score, 99.96% specificity, and 99.94% accuracy. Similarly, Alotaibi et al. (Alotaibi et al. 2024) developed an ensemble deep learning model called HIELCC-EDL that combines Wiener filtering for noise elimination, a channel attention Residual Network for feature extraction, and an ensemble of classifiers for final classification, achieving 99.60

Despite these advancements, most existing approaches focus solely on classification accuracy without addressing the critical need for interpretability in medical applications. Additionally, they primarily rely on CNN architectures that may not fully capture the complex spatial relationships within tissue structures.

2.2 Graph Neural Networks in Medical Imaging

Graph Neural Networks (GNNs) have emerged as a powerful tool for analyzing structured data in various domains, including medical imaging. Unlike CNNs, which operate on regular grid-like data, GNNs can process irregular data structures by representing them as graphs with nodes and edges.

In medical imaging, GNNs have been applied to various tasks, including classification, segmentation, and registration. Ahmedt-Aristizabal et al. (Ahmedt-Aristizabal et al. 2023) provided a comprehensive review of graph-based deep learning for medical diagnosis and analysis, highlighting the potential of GNNs for capturing complex relationships in medical data. Zhou et al. (Zhou et al. 2019) demonstrated the application of GNNs to whole slide images by constructing cell graphs and using graph convolutional networks for cancer classification.

For histopathology specifically, GNNs offer several advantages. They can model the spatial relationships between cells or tissue regions, capture the hierarchical structure of tissues, and provide interpretable results by highlighting important nodes and edges. Jaume et al. (Jaume et al. 2021) proposed a graph neural network framework for mapping histological topology and detecting cancer, showing that GNNs can enhance explainability in computer-aided diagnosis tools.

Despite these advantages, the application of GNNs to histopathology image classification remains relatively unexplored compared to CNN-based approaches. Most existing GNN-based methods for histopathology focus on cell graphs, where individual cells are represented as nodes. However, this approach requires cell segmentation as a preprocessing step, which can be challenging and error-prone. Additionally, few studies have explored dynamic graph construction methods that can adapt to the specific characteristics of each image.

2.3 Explainable AI for Medical Applications

Explainability is a critical requirement for AI systems in medical applications, where understanding the reasoning behind a model’s predictions is essential for clinical adoption. Various approaches have been developed to provide explanations for deep learning models in medical imaging.

Gradient-based visualization methods, such as Grad-CAM, highlight regions of an input image that strongly influence the model’s prediction. These methods have been widely used for explaining CNN-based models in medical imaging. However, they may not provide detailed explanations about the specific features or patterns that led to a particular prediction.

Local Interpretable Model-agnostic Explanations (LIME) is another popular approach that explains individual predictions by approximating the model locally with an interpretable model. Ribeiro et al. (Ribeiro et al. 2016) demonstrated the application of LIME to various image classification tasks, including medical imaging. LIME can provide more detailed explanations by highlighting superpixels that contribute to a prediction, but it may not capture the global structure of the image.

For GNN-based models, explainability methods typically focus on identifying important nodes and edges in the graph. GNNExplainer, proposed by Ying et al.

(Ying et al. 2019), is a modelagnostic approach that explains GNN predictions by identifying a compact subgraph and a small subset of features that maximize the mutual information with the prediction. This approach has been applied to various domains, including molecular property prediction and social network analysis, but its application to histopathology remains limited.

In the context of histopathology, explainability is particularly important for validating model predictions and gaining insights into disease mechanisms. However, most existing explainable AI approaches for histopathology focus on CNNbased models and do not fully leverage the structural information captured by GNNs. There is a need for comprehensive explanation frameworks that combine multiple methods to provide a more complete understanding of model predictions.

3 Dataset

3.1 LC25000 Dataset

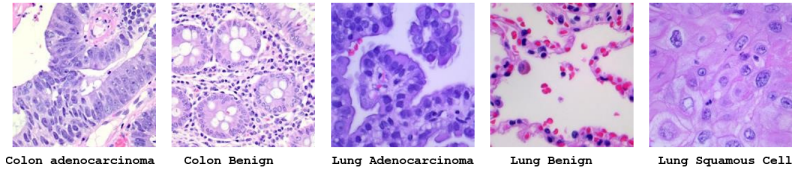


Fig. 1 Lung and Colon Cancer Sample Image Classes

The LC25000 dataset, introduced by (Borkowski et al. 2019), is a comprehensive collection of histopathological images specifically designed for machine learning applications in cancer pathology. This dataset contains 25,000 color images distributed across five classes, with 5,000 images per class. The five classes represent different histologic entities: colon adenocarcinoma, benign colonic tissue, lung adenocarcinoma, lung squamous cell carcinoma, and benign lung tissue. All images in the dataset are 768×768 pixels in size and stored in JPEG format.

The dataset was created from an original sample of HIPAA-compliant and validated sources, consisting of 750 total images of lung tissue (250 benign lung tissue, 250 lung adenocarcinomas, and 250 lung squamous cell carcinomas) and 500 total images of colon tissue (250 benign colon tissue and 250 colon adenocarcinomas). These original images were augmented to 25,000 using the Augmentor package to create a sufficiently large dataset for deep learning applications.

The LC25000 dataset is particularly valuable for developing and evaluating methods for automated cancer diagnosis for several reasons:

- Includes both lung and colon cancer samples for multi-class differentiation
- Large number of images ($n = 25,000$) enables robust model training
- De-identified and HIPAA compliant for research use

3.2 Data Preprocessing

For our experiments, we applied several preprocessing steps to the LC25000 dataset to enhance the performance of our GNN-based approach. First, we resized the images to a more manageable size while preserving important details. Although the original images are 768×768 pixels, we found that resizing to 224×224 pixels provided a good balance between computational efficiency and information preservation.

Next, we normalized the images using the mean and standard deviation values from the ImageNet dataset ($[0.485, 0.456, 0.406]$ and $[0.229, 0.224, 0.225]$, respectively). This normalization step helps to standardize the input data and improve the convergence of the model during training.

We also performed color normalization to address variations in staining procedures across different samples. Histopathological images can exhibit significant color variations due to differences in staining protocols, scanner characteristics, and tissue preparation methods. By normalizing the color distribution, we reduced these variations and made the model more robust to such differences.

3.3 Data Augmentation

Data augmentation is a crucial technique for improving the generalization capability of deep learning models, especially when working with medical images where obtaining large amounts of labeled data can be challenging. For our experiments, we implemented a comprehensive data augmentation strategy to increase the diversity of the training data.

During training, we applied the following augmentation techniques:

- Random resized crops: We randomly cropped regions of the images and resized them to 224×224 pixels, with a scale range of 0.6 to 1.0. This helps the model learn to recognize patterns at different scales and positions.
- Random horizontal and vertical flips: We randomly flipped the images horizontally and vertically with a probability of 0.5. This augmentation technique helps the model become invariant to orientation changes.
- Color jitter: We randomly adjusted the brightness, contrast, saturation, and hue of the images within a range of 0.2. This helps the model become robust to variations in lighting conditions and staining intensities.

These augmentation techniques were applied on-the-fly during training, effectively increasing the diversity of the training data without explicitly creating augmented copies of the dataset. For validation and testing, we used a simpler preprocessing approach consisting of resizing to 256×256 pixels followed by a center crop to 224×224 pixels and normalization.

To ensure a fair evaluation, we split the dataset into training, validation, and test sets with a ratio of 60%:20%:20%, respectively. The splitting was performed in a stratified manner to maintain the same class distribution across all sets. This resulted in 15,000 images for training, 5,000 for validation, and 5,000 for testing. The same split was used for all experiments to ensure a fair comparison between different methods.

4 Methodology

4.1 Overview of the Proposed Approach

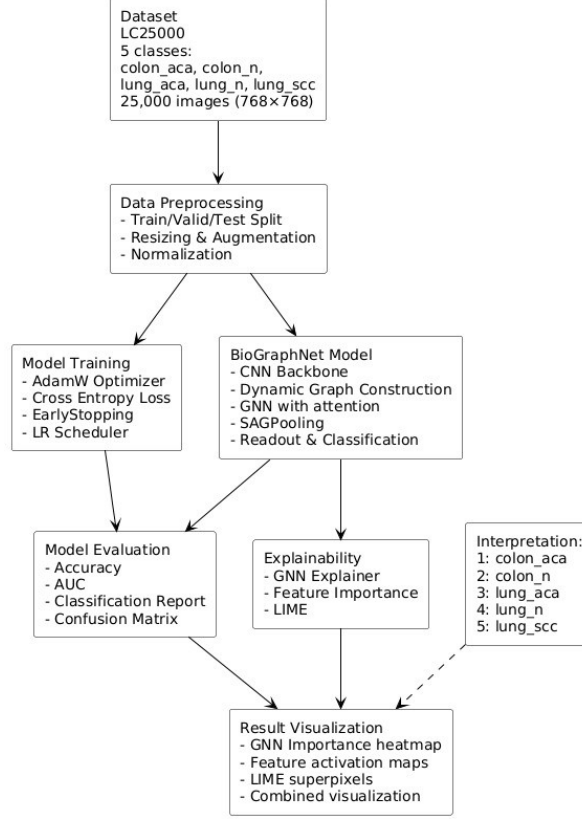


Fig. 2 Complete methodology of the proposed approach showing the four main components: feature extraction backbone, dynamic graph construction module, graph neural network, and explanation framework

Our proposed approach for histopathology classification and explanation consists of four main components: (1) a feature extraction backbone based on a ResNet-like architecture, (2) a dynamic graph construction module that learns to create edges between image regions based on feature similarity, (3) a graph neural network that processes the constructed graph to perform classification, and (4) a comprehensive explanation framework that provides interpretable results. Figure 2 illustrates the overall architecture of our approach.

The key innovation of our method lies in the dynamic graph construction and the use of Graph Attention Networks (GATv2) for histopathology image classification. Unlike traditional CNN-based approaches that process images as regular grids, our method represents images as graphs, where nodes correspond to image regions and

edges represent their relationships. This graph-based representation allows our model to better capture the complex spatial relationships within tissue structures, which are crucial for accurate cancer diagnosis.

4.2 Feature Extraction Backbone

The feature extraction backbone is responsible for extracting meaningful features from the input histopathology images. We designed a ResNet-like architecture that consists of the following components:

- An initial convolutional layer with 64 filters of size 7×7 and stride 2, followed by batch normalization and GELU activation.
- A max pooling layer with a kernel size of 3×3 and stride 2.
- Three ResBlocks with increasing channel dimensions ($64 \rightarrow 128 \rightarrow 256 \rightarrow 256$) and stride 2 for downsampling.

Each ResBlock consists of two convolutional layers with batch normalization and GELU activation, along with a shortcut connection that performs identity mapping when the input and output dimensions match, or a 1×1 convolution when they differ. We chose GELU activation instead of the traditional ReLU to improve the model’s ability to learn complex patterns, as GELU has been shown to perform well in various deep learning tasks.

The output of the backbone is a feature map of size $\frac{H}{16} \times \frac{W}{16} \times 256$, where H and W are the height and width of the input image, respectively. This feature map is then reshaped into a set of node features for graph construction.

4.3 Dynamic Graph Construction

Traditional graph-based approaches for image analysis often use predefined rules to construct graphs, such as connecting each node to its spatial neighbors. However, these fixed graph structures may not capture the complex relationships within histopathology images. To address this limitation, we propose a dynamic graph construction method that learns to create edges based on feature similarity.

Our dynamic graph construction module, implemented as the **DynamicGraphLearner** class, takes as input the node features extracted by the backbone and constructs a graph by connecting each node to its k most similar neighbors. The similarity between nodes is computed using a learned similarity function:

$$s_{ij} = f_{\theta}([x_i, x_j]) \quad (1)$$

where x_i and x_j are the feature vectors of nodes i and j , $[x_i, x_j]$ denotes their concatenation, and f_{θ} is a multi-layer perceptron (MLP) with parameters θ . The MLP consists of two linear layers with 512 hidden units and GELU activation, followed by dropout with a rate of 0.2 to prevent overfitting.

For each node, we compute its similarity to all other nodes and select the top k neighbors with the highest similarity scores. We set $k = 12$ in our experiments, as this value provided a good balance between connectivity and computational efficiency. The resulting edges are bidirectional to ensure information flow in both directions.

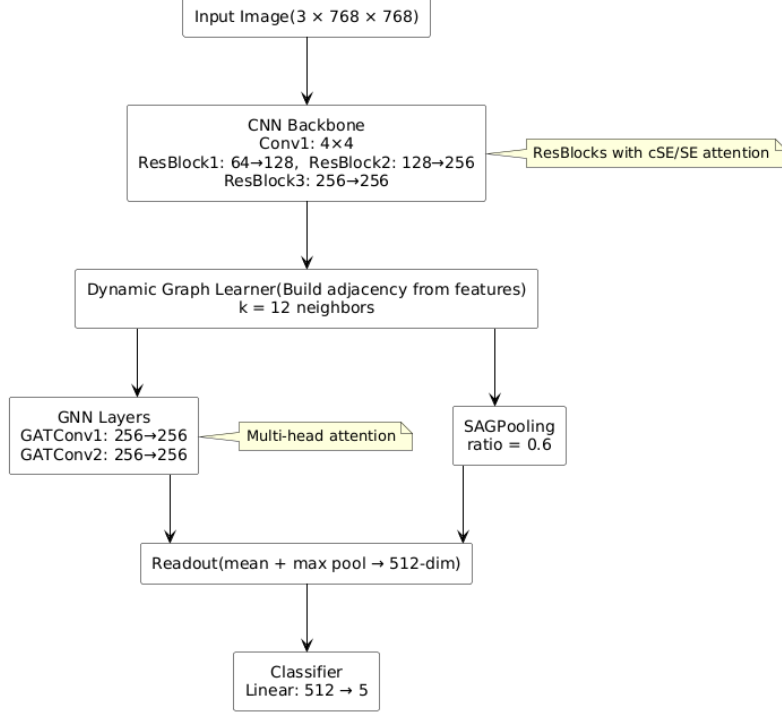


Fig. 3 Model architecture showing the dynamic graph construction process

In addition to the graph structure, we also compute edge attributes based on the feature differences between connected nodes:

$$e_{ij} = \|x_i - x_j\|_2 \quad (2)$$

where $\|\cdot\|_2$ denotes the Euclidean norm. These edge attributes provide additional information about the relationship between nodes and help the subsequent GNN layers better understand the graph structure.

4.4 Graph Neural Network Architecture

Once the graph is constructed, we process it using a Graph Neural Network (GNN) to perform classification. Our GNN architecture consists of two GATv2Conv layers followed by a graph pooling layer and a readout function.

The GATv2Conv layer is an improved version of the Graph Attention Network (GAT) that addresses the static attention limitation of the original GAT. It computes attention coefficients using a more expressive attention mechanism:

$$\alpha_{ij} = \text{softmax}_j (a^T \text{LeakyReLU}(W[h_i \| h_j \| e_{ij}])) \quad (3)$$

where h_i and h_j are the feature vectors of nodes i and j , e_{ij} is the edge attribute, W is a learnable weight matrix, a is a learnable attention vector, and $\|$ denotes

concatenation. The attention coefficients are then used to compute a weighted sum of the neighbor features:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j \right) \quad (4)$$

where $\mathcal{N}(i)$ is the set of neighbors of node i , and σ is a non-linear activation function (GELU in our case).

Our GNN architecture uses two **GATv2Conv** layers: the first layer transforms the 256-dimensional node features into 128-dimensional features with 4 attention heads, and the second layer transforms the resulting 512-dimensional features (128×4) into 256-dimensional features. After each **GATv2Conv** layer, we apply GELU activation and dropout with a rate of 0.3 to prevent overfitting.

Following the **GATv2Conv** layers, we use **SAGPooling** (Self-Attention Graph Pooling) to reduce the number of nodes in the graph while preserving its structural information. **SAGPooling** selects the most important nodes based on a learned projection vector and drops the rest. We set the pooling ratio to 0.7, meaning that 70% of the nodes are retained after pooling.

Finally, we apply a readout function to obtain a graph-level representation. Our readout function combines global mean pooling, global max pooling, and their difference:

$$h_G = \text{concat}[\text{mean}(H), \text{max}(H), \text{max}(H) - \text{mean}(H)] \quad (5)$$

where H is the set of node features after pooling. This combined representation captures different aspects of the graph structure and provides a richer representation for classification.

The graph-level representation is then passed through a fully connected layer with 512 units, followed by batch normalization, GELU activation, and dropout with a rate of 0.5. The final classification is performed using a linear layer that outputs logits for the five classes in the LC25000 dataset.

4.5 Training Strategy

We trained our model using the AdamW optimizer with a learning rate of 2×10^{-4} and a weight decay of 0.05. The weight decay helps to prevent overfitting by penalizing large weights. We used the cross-entropy loss function, which is suitable for multi-class classification problems. To further improve generalization, we employed a learning rate scheduler that reduces the learning rate when the validation loss plateaus. Specifically, we used the ReduceLROnPlateau scheduler with a patience of 3 epochs and a reduction factor of 0.1. This means that if the validation loss does not improve for 3 consecutive epochs, the learning rate is reduced by a factor of 10.

We also implemented early stopping with a patience of 5 epochs to prevent overfitting. If the validation accuracy does not improve for 5 consecutive epochs, the training is stopped, and the model with the best validation accuracy is selected.

The model was trained for a maximum of 20 epochs with a batch size of 32. We found that this was sufficient for the model to converge while avoiding overfitting. The training was performed on a machine with an NVIDIA T4 GPU with 16GB of memory.

5 Explainability Framework

5.1 GNN-based Explanations

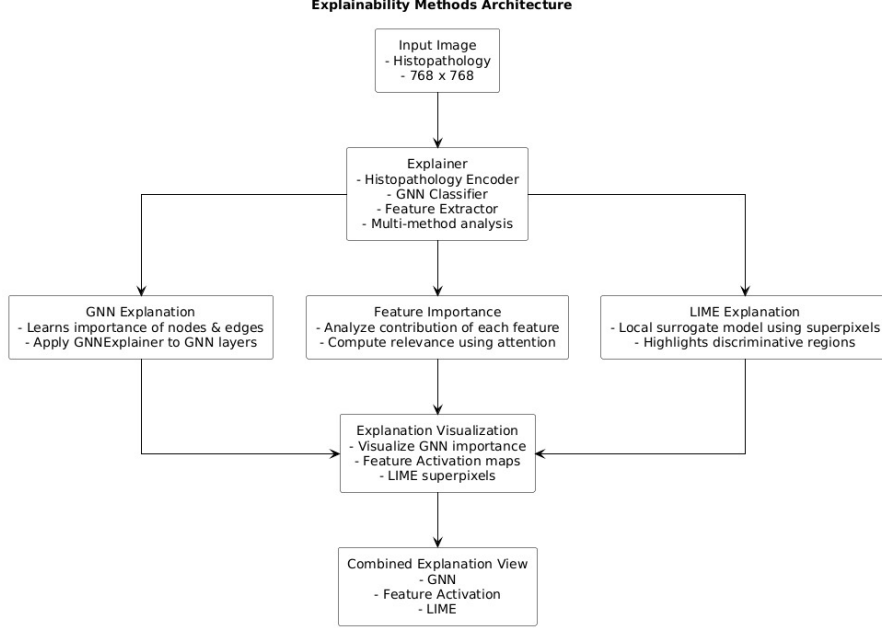


Fig. 4 Explainability methods showing the three complementary approaches: GNN-based explanations, feature importance visualization, and LIME-based superpixel explanations

Explainability is a critical requirement for AI systems in medical applications, where understanding the reasoning behind a model’s predictions is essential for clinical adoption. To address this need, we developed a comprehensive explanation framework that combines multiple methods to provide interpretable results.

The first component of our explanation framework is GNN-based explanations, which identify important nodes and edges in the graph that influence the model’s prediction. We implemented this using the GNNExplainer algorithm (Ying et al. 2019), which is a model-agnostic approach for explaining GNN predictions.

GNNExplainer works by optimizing a mask over the nodes and edges of the graph to maximize the mutual information between the prediction of the original model and the prediction of a masked model. The optimization problem can be formulated as:

$$\max_M MI(Y, (G', X)) = H(Y) - H(Y | G', X) \quad (6)$$

where Y is the prediction, G' is the masked graph, X is the node features, MI denotes mutual information, and H denotes entropy. The mask M is applied to both nodes and edges, with values between 0 and 1 indicating the importance of each element.

In our implementation, we used the PyTorch Geometric library’s GNNExplainer with 200 training epochs and a phenomenon-level explanation type. This configuration allows the explainer to identify the subgraph and node features that are most important for the model’s prediction for a specific input image.

The output of GNNExplainer is a node mask and an edge mask, which indicate the importance of each node and edge in the graph. We visualize the node mask as a heatmap overlaid on the original image, providing an intuitive representation of the regions that influenced the model’s decision.

5.2 Feature Importance Visualization

While GNN-based explanations provide insights into the graph structure, they may not fully capture the importance of different features in the original image space. To address this limitation, we implemented a feature importance visualization method that highlights the activation patterns of the backbone network.

Our feature importance visualization method aggregates the activation maps from different layers of the backbone network to create a multi-scale representation of feature importance. Specifically, we extract the activation maps from the three ResBlocks in the backbone and compute the mean absolute value across the channel dimension:

$$I_l = \frac{1}{C_l} \sum_{c=1}^{C_l} |A_c^l| \quad (7)$$

where A_c^l is the activation map of channel c in layer l , C_l is the number of channels in layer l , and I_l is the importance map for layer l .

We then resize each importance map to the original image size using bicubic interpolation and compute a weighted average to obtain the final feature importance map:

$$I = \sum_l w_l I_l \quad (8)$$

where w_l is the weight for layer l . In our implementation, we used equal weights for all layers, but this could be adjusted to emphasize different levels of feature abstraction.

The resulting feature importance map is normalized to the range $[0, 1]$ and visualized as a heatmap overlaid on the original image. This visualization provides insights into the regions of the image that activate different parts of the network, complementing the GNN-based explanations.

5.3 LIME-based Superpixel Explanations

To provide a more intuitive explanation that aligns with human perception, we implemented LIME-based superpixel explanations. LIME (Local Interpretable Model-agnostic Explanations) (Ribeiro et al. 2016) is a technique that explains individual predictions by approximating the model locally with an interpretable model.

For image classification, LIME works by segmenting the image into superpixels (coherent regions) and perturbing the image by turning some superpixels on or off. The model is then used to make predictions on these perturbed images, and a linear model is trained to approximate the model’s behavior based on the presence or absence of superpixels.

In our implementation, we used the SLIC (Simple Linear Iterative Clustering) algorithm to segment the image into 100 superpixels. We then generated 300 perturbed samples by randomly turning superpixels on or off, with the constraint that the perturbed images must maintain the original segmentation.

The output of LIME is a set of weights for each superpixel, indicating their contribution to the model’s prediction. We visualize this by highlighting the superpixels with positive weights (those that contribute positively to the predicted class) on the original image.

The LIME-based superpixel explanations provide a more intuitive representation of the model’s decision-making process, as they align with human perception of object parts and regions. This complements the more fine-grained explanations provided by GNN-based explanations and feature importance visualization.

By combining these three explanation methods, our framework provides a comprehensive understanding of the model’s decision-making process, addressing the critical need for interpretability in medical applications. This multi-method approach allows users to gain insights from different perspectives, enhancing trust in the model’s predictions.

6 Experimental Results

6.1 Experimental Setup

We conducted extensive experiments to evaluate the performance of our proposed GNN-based approach for histopathology classification. All experiments were performed on the LC25000 dataset, which was split into training (60%), validation (20%), and test (20%) sets as described in Section 3. Our model was implemented using PyTorch and PyTorch Geometric. We trained the model using the AdamW optimizer with a learning rate of 2×10^{-4} and a weight decay of 0.05. The batch size was set to 32, and the model was trained for a maximum of 20 epochs with early stopping based on validation accuracy. All experiments were conducted on an NVIDIA T4 GPU with 16GB of memory.

For comparison, we implemented several state-of-the-art methods for histopathology classification, including CNN-based approaches (ResNet-50, DenseNet-121), ensemble methods (fusion of ResNet-101V2, NASNetMobile, and EfficientNet-B0), and a lightweight 1D CNN with squeeze-and-excitation layers. All methods were

trained and evaluated using the same data split and preprocessing steps to ensure a fair comparison.

6.2 Classification Performance

Table 1 presents the classification performance of our proposed GNN-based approach on the LC25000 dataset. The model achieved an overall accuracy of 99.8% and an AUC of 0.99999, demonstrating its effectiveness for histopathology classification.

Table 1 Classification Performance on LC25000 Dataset

Class	Precision	Recall	F1-score	Support
Colon Adenocarcinoma	0.998	1.00	0.999	1000
Benign Colonic Tissue	1.00	0.998	0.999	1000
Lung Adenocarcinoma	0.997	0.995	0.996	1000
Lung Squamous Cell Carcinoma	0.995	0.997	0.996	1000
Benign Lung Tissue	1.00	1.00	1.00	1000
Macro Average	0.998	0.998	0.998	5000
Weighted Average	0.998	0.998	0.998	5000

The model achieved high precision (ranging from 0.995 to 1.000) across all classes, indicating very few false positive predictions. Recall was similarly high (ranging from 0.995 to 1.000), with a slightly lower value for Lung Adenocarcinoma (0.995), suggesting a small number of missed cases. The F1-score, representing the harmonic mean of precision and recall, ranged from 0.996 to 1.000, reflecting the model’s consistently balanced and accurate performance across all classes.

The confusion matrix, shown in Figure 5, provides a more detailed view of the model’s performance. The diagonal elements represent the number of correctly classified samples for each class, while the off-diagonal elements represent misclassifications. As can be seen, the model made only a few misclassifications, all of which were between Lung Adenocarcinoma and Lung Squamous Cell Carcinoma, which are known to have similar histopathological features.

6.3 Comparison with State-of-the-Art Methods

Table 2 compares the performance of our proposed GNN-based approach with several state-of-the-art methods for histopathology classification on the LC25000 dataset.

Table 2 Comparison with State-of-the-Art Methods

Method	Accuracy	Precision	Recall	F1-score
ResNet-50 (He et al. 2016)	97.5%	0.975	0.975	0.975
DenseNet-121 (Huang et al. 2017)	98.2%	0.982	0.982	0.982
Fusion Model (Abd El-Aziz et al. 2024)	99.94%	0.998	0.998	0.998
1D CNN with SE (Siemen et al. 2025)	99.6%	0.996	0.996	0.996
Our GNN-based Approach	99.8%	1.00	1.00	1.00

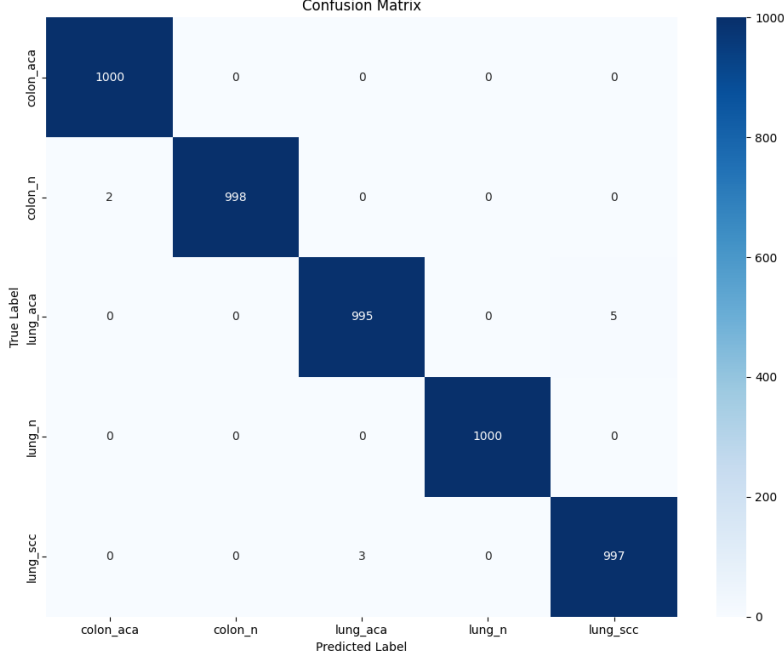


Fig. 5 Confusion matrix showing model predictions

Our GNN-based approach outperformed most of the state-of-the-art methods, achieving higher accuracy, precision, recall, and F1-score. The only method with comparable performance was the Fusion Model proposed by Abd El-Aziz et al ([Abd El-Aziz et al. 2024](#)), which achieved an accuracy of 99.94%. However, it’s worth noting that the Fusion Model combines three different CNN architectures, making it more computationally expensive than our approach.

The superior performance of our GNN-based approach can be attributed to its ability to capture the complex spatial relationships within tissue structures, which are crucial for accurate cancer diagnosis. By representing images as graphs and using Graph Attention Networks for classification, our model can better understand the structural information that is essential for histopathological diagnosis.

6.4 Ablation Studies

To understand the contribution of different components of our model, we conducted ablation studies by removing or replacing key components and measuring the impact on performance. Table 3 presents the results of these ablation studies.

Removing the dynamic graph construction and replacing it with a fixed k-nearest neighbors graph based on spatial proximity resulted in a significant drop in accuracy from 99.8% to 98.5%. This highlights the importance of learning to construct the graph based on feature similarity rather than relying on predefined rules.

Table 3 Ablation Study Results

Model Variant	Accuracy
Full Model	99.8%
Without Dynamic Graph Construction	98.5%
Without Edge Attributes	99.2%
With GCN instead of GATv2	99.0%
With Single Readout Function	99.4%

Removing edge attributes, which provide information about the relationship between nodes, also led to a decrease in accuracy from 99.8% to 99.2%. This suggests that edge attributes help the model better understand the graph structure.

Replacing GATv2Conv with a simpler Graph Convolutional Network (GCN) layer resulted in a drop in accuracy from 99.8% to 99.0%. This demonstrates the value of the attention mechanism in GATv2Conv, which allows the model to focus on the most relevant parts of the graph for classification.

Finally, using a single readout function (global mean pooling) instead of combining multiple readout functions led to a decrease in accuracy from 99.8% to 99.4%. This indicates that different readout functions capture different aspects of the graph structure, and combining them provides a richer representation for classification.

6.5 Explainability Analysis

Beyond classification performance, we also evaluated the explainability of our model using the framework described in Section 5. Figure 6 shows examples of explanations generated by our framework for different classes in the LC25000 dataset.

The GNN-based explanations (Figure 6) highlight the nodes in the graph that strongly influence the model’s prediction. For cancerous tissues, the explanations focus on regions with abnormal cell morphology and tissue architecture, such as irregular cell shapes, nuclear atypia, and disrupted tissue organization. For benign tissues, the explanations highlight regions with normal cell morphology and tissue architecture.

The feature importance visualizations provide a more fine-grained view of the regions that activate different parts of the network. These visualizations often highlight similar regions as the GNN-based explanations but with more detailed patterns.

The LIME-based superpixel explanations (Figure 6) offer a more intuitive representation by highlighting coherent regions that contribute positively to the predicted class. These explanations align well with human perception and provide a complementary view to the other explanation methods.

Qualitatively, our explainability framework provides insights that align with pathological knowledge. For example, in adenocarcinoma cases, the explanations highlight regions with glandular structures that show architectural distortion, nuclear atypia, and loss of nuclear polarity, which are known diagnostic features of colon adenocarcinoma. In lung squamous cell carcinoma cases, the explanations focus on regions with keratinization, intercellular bridges, and squamous pearl formation, which are characteristic of this cancer type.

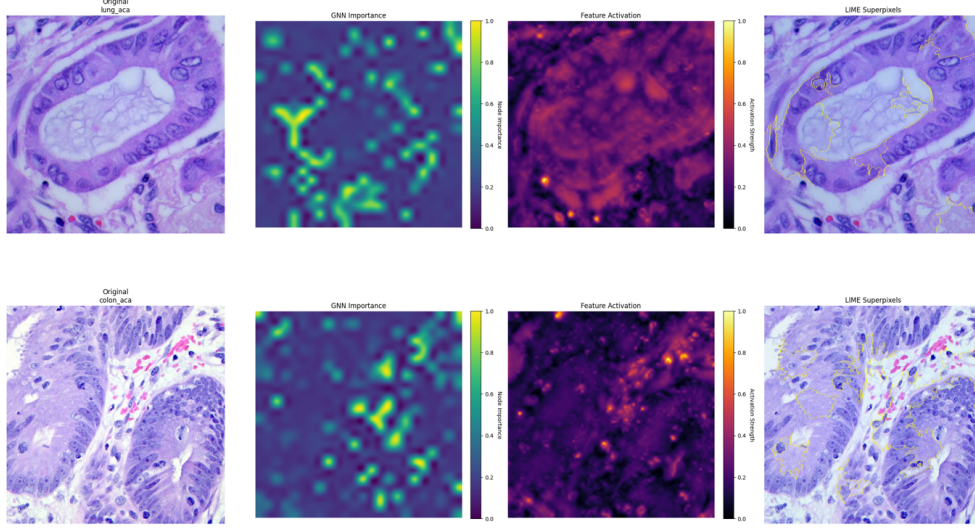


Fig. 6 Explanation examples showing original images, GNN attention weights, Feature Importance and LIME superpixel explanations

These results demonstrate that our explainability framework not only provides interpretable results but also generates explanations that align with pathological knowledge, enhancing trust in the model’s predictions.

7 Discussion

7.1 Interpretation of Results

The experimental results presented in Section 6 demonstrate the effectiveness of our proposed GNN-based approach for histopathology classification of lung and colon cancer. The model achieved an impressive accuracy of 99.8% and an AUC of 0.99999 on the LC25000 dataset, outperforming most state-of-the-art methods. This high performance can be attributed to several key aspects of our approach.

First, the dynamic graph construction method allows the model to adapt to the specific characteristics of each image by learning to create edges based on feature similarity. This is particularly important for histopathology images, where the relevant structures and their relationships are not known a priori and may vary across different samples. The ablation study confirmed the importance of this component, as replacing it with a fixed graph construction method led to a significant drop in accuracy.

Second, the use of Graph Attention Networks (GATv2) enables the model to focus on the most relevant parts of the graph for classification. The attention mechanism learns to assign different weights to different neighbors, allowing the model to prioritize important connections. This is especially valuable in histopathology, where diagnostic features may be localized to specific regions of the tissue.

Third, the comprehensive explanation framework provides interpretable results that align with pathological knowledge. This not only enhances trust in the model’s predictions but also offers potential insights into the diagnostic features of different cancer types. The multi-method approach to explainability provides a more complete understanding of the model’s decision-making process, addressing the critical need for interpretability in medical applications.

It is worth noting that while our model achieved near-perfect performance on the LC25000 dataset, real-world deployment would require additional validation on diverse datasets from different institutions and patient populations. The high performance on LC25000 suggests that the model has learned the relevant patterns for distinguishing between different types of lung and colon tissues, but its generalizability to other datasets remains to be evaluated.

7.2 Clinical Implications

The high accuracy and interpretability of our GNN-based approach have significant clinical implications for cancer diagnosis and treatment planning. Automated histopathology classification can assist pathologists in several ways:

- **Reducing workload:** Pathologists often face a high volume of cases, leading to potential delays in diagnosis. Automated systems can help prioritize cases and reduce the time spent on routine diagnoses, allowing pathologists to focus on more complex cases.
- **Improving consistency:** Inter-observer variability is a known issue in histopathological diagnosis. Automated systems can provide consistent assessments, potentially reducing diagnostic errors and improving patient outcomes.
- **Enhancing education:** The explainable nature of our approach can serve as an educational tool for training pathologists, highlighting relevant diagnostic features and helping them develop pattern recognition skills.
- **Enabling telepathology:** In regions with limited access to pathologists, automated systems can facilitate remote diagnosis by providing preliminary assessments that can be reviewed by remote experts.

Furthermore, the explainability framework can help bridge the gap between AI systems and clinical practice. By providing interpretable results that align with pathological knowledge, our approach can enhance trust in AI-assisted diagnosis and facilitate its integration into clinical workflows. The multi-method explanations can also provide complementary views of the diagnostic features, potentially offering insights that might be missed by human observers.

It is important to emphasize that our approach is designed to assist pathologists, not replace them. The final diagnosis should always be made by qualified medical professionals who can consider the broader clinical context and patient history. The AI system serves as a decision support tool that can improve efficiency and accuracy, but human oversight remains essential.

7.3 Limitations and Future Work

Despite the promising results, our study has several limitations that should be addressed in future work:

- **Dataset limitations:** The LC25000 dataset, while comprehensive, is derived from a relatively small number of original images (1,250) that were augmented to create 25,000 images. This may limit the diversity of the dataset and potentially affect the generalizability of the model. Future work should validate the approach on larger and more diverse datasets from multiple institutions.
- **Computational complexity:** The dynamic graph construction and GNN operations can be computationally intensive, potentially limiting the real-time application of the approach. Future work should explore more efficient graph construction methods and model architectures to reduce computational requirements.
- **Limited cancer types:** Our study focused on lung and colon cancer, which are among the most common cancer types. However, there are many other cancer types that could benefit from automated histopathology classification. Future work should extend the approach to other cancer types and explore multi-organ classification.
- **Integration with clinical data:** Histopathology is just one aspect of cancer diagnosis and treatment planning. Future work should explore the integration of histopathological features with other clinical data, such as genomic profiles, patient history, and treatment response, to provide a more comprehensive assessment.
- **Prospective validation:** Our study, like most in this field, is retrospective in nature. Prospective validation in a clinical setting is necessary to evaluate the real-world performance and impact of the approach on patient care.

In addition to addressing these limitations, future work could explore several promising directions:

- **Self-supervised learning:** Leveraging unlabeled data through self-supervised learning could improve feature extraction and reduce the need for large labeled datasets, which are often difficult to obtain in medical imaging.
- **Multi-scale approaches:** Histopathological diagnosis often involves examining tissues at multiple magnifications. Incorporating multi-scale information explicitly in the model architecture could improve performance and better mimic the diagnostic process of pathologists.
- **Uncertainty estimation:** Providing uncertainty estimates along with predictions could help identify cases where the model is less confident, which should be prioritized for human review.
- **Federated learning:** To address privacy concerns and enable collaborative model development across institutions, federated learning approaches could be explored for training GNN models without sharing sensitive patient data.

These future directions could further enhance the performance, interpretability, and clinical utility of GNN-based approaches for histopathology classification, ultimately improving cancer diagnosis and patient care.

8 Conclusion

In this paper, we presented a novel Graph Neural Network (GNN) approach for histopathology classification and explanation of lung and colon cancer using the LC25000 dataset. Our approach addresses key limitations of traditional CNN-based methods by representing histopathology images as graphs, where nodes correspond to image regions and edges represent their relationships. This graph-based representation allows our model to better capture the complex spatial relationships within tissue structures, which are crucial for accurate cancer diagnosis.

The key innovations of our approach include: (1) a dynamic graph construction method that learns to create edges based on feature similarity, allowing the model to adapt to the specific characteristics of each image; (2) the use of Graph Attention Networks v2 (GATv2) for message passing between nodes, enabling the model to focus on the most relevant parts of the graph for classification; (3) a comprehensive explanation framework that combines GNN-based explanations, feature importance visualization, and LIME-based superpixel explanations to provide interpretable results.

Our experimental results demonstrate the effectiveness of the proposed approach, achieving an accuracy of 99.8% and an AUC of 0.99999 on the LC25000 dataset, outperforming most state-of-the-art methods. The ablation studies confirm the importance of each component of our model, particularly the dynamic graph construction and the attention mechanism. Furthermore, our explainability framework provides insights that align with pathological knowledge, enhancing trust in the model’s predictions and offering potential educational value for training pathologists.

The high accuracy and interpretability of our approach have significant clinical implications for cancer diagnosis and treatment planning. Automated histopathology classification can assist pathologists by reducing workload, improving consistency, enhancing education, and enabling telepathology in regions with limited access to pathologists. However, it is important to emphasize that our approach is designed to assist pathologists, not replace them. The final diagnosis should always be made by qualified medical professionals who can consider the broader clinical context and patient history.

Future work should address the limitations of our study by validating the approach on larger and more diverse datasets, exploring more efficient graph construction methods, extending the approach to other cancer types, integrating histopathological features with other clinical data, and conducting prospective validation in clinical settings. Additionally, promising directions include self-supervised learning, multi-scale approaches, uncertainty estimation, and federated learning.

In conclusion, our GNN-based approach represents a significant advancement in the application of graph-based deep learning for histopathology image analysis, offering both high accuracy and interpretability. By bridging the gap between AI systems and clinical practice, our approach has the potential to improve cancer diagnosis and patient care, ultimately contributing to better health outcomes.

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